Matrices

Introduction

Solution 2

A **matrix** (plural: matrices) is an array of numbers arranged in rows and columns.

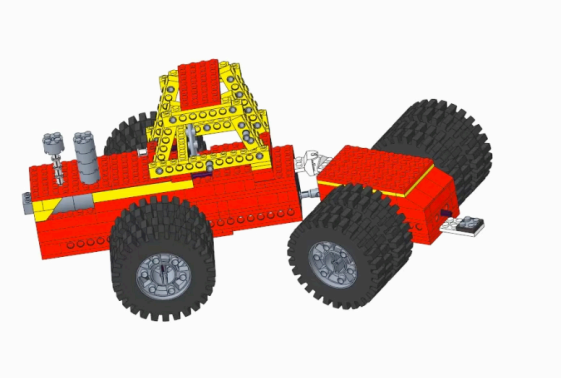
$$\begin{pmatrix} 1 & 0 & -2 \\ 3 & 3 & 0 \end{pmatrix}$$

A matrix organises several values into one structure.

Why use matrices?

They represent linear transformations, including rotations, reflections and enlargements, and help solve simultaneous equations.

In 3D graphics, matrices control transformations and camera angles.



Example of matrices

Solution 3

In computer science, matrices are also called arrays. This example records the top five teams in a Premier League table.

Columns: matches played, won, drawn, lost, goals for, goals against, goal difference, points.

$$\begin{pmatrix} 26 & 20 & 3 & 3 & 59 & 25 & 34 & 63 \\ 27 & 19 & 4 & 4 & 67 & 25 & 42 & 61 \\ 25 & 15 & 4 & 6 & 41 & 35 & 6 & 49 \\ 27 & 15 & 3 & 9 & 49 & 37 & 12 & 48 \\ 26 & 12 & 6 & 8 & 47 & 29 & 18 & 42 \end{pmatrix}$$

There are **5 rows** and **8 columns**: a 5×8 matrix.

Dimension of Matrices

Solution 4

The dimension of a matrix is its number of rows followed by its number of columns.

$\begin{pmatrix} 1 & 3 & -7 \\ 4 & 0 & 5 \end{pmatrix}$

2×3

$\begin{pmatrix} 1 \\ 6 \\ -3 \end{pmatrix}$

3×1

$\begin{pmatrix} 1 & 6 & 0 \end{pmatrix}$

1×3

Matrix Notation

Solution 5

Matrices may use square or curved brackets. A matrix with one column is a column vector; a matrix with one row is a row vector.

$$\begin{pmatrix} 1 & 3 & -7 \\ 4 & 0 & 5 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 6 \\ -3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 6 & 0 \end{pmatrix}$$

Matrix variables use **bold capital letters**; vector variables use bold lowercase letters.

$$\mathbf{A} = \begin{pmatrix} 1 \\ 6 \\ -3 \end{pmatrix}$$

$$\mathbf{C} = \mathbf{P}^{-1}\mathbf{T}\mathbf{P}$$

Matrix Fundamentals: Adding/Subtracting Matrices

Solution 6

Arsenal and Liverpool draw 2–2. Add the match-result matrix to the original standings. Only matrices of the same dimensions can be added or subtracted.

Original standings

$$\begin{pmatrix} 26 & 20 & 3 & 3 & 59 & 25 & 34 & 63 \\ 27 & 19 & 4 & 4 & 67 & 25 & 42 & 61 \\ 25 & 15 & 4 & 6 & 41 & 35 & 6 & 49 \\ 27 & 15 & 3 & 9 & 49 & 37 & 12 & 48 \\ 26 & 12 & 6 & 8 & 47 & 29 & 18 & 42 \end{pmatrix}$$

Match result

$$\begin{pmatrix} 1 & 0 & 1 & 0 & 2 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 2 & 2 & 0 & 1 \end{pmatrix}$$

Updated table

New standings

$$\begin{pmatrix} 27 & 20 & 4 & 3 & 61 & 27 & 34 & 64 \\ 27 & 19 & 4 & 4 & 67 & 25 & 42 & 61 \\ 25 & 15 & 4 & 6 & 41 & 35 & 6 & 49 \\ 27 & 15 & 3 & 9 & 49 & 37 & 12 & 48 \\ 27 & 12 & 7 & 8 & 49 & 31 & 18 & 43 \end{pmatrix}$$

Now you try:

Solution 7

Add or subtract corresponding elements:

$$\begin{pmatrix} 1 & 3 & -7 \\ 4 & 0 & 5 \end{pmatrix} + \begin{pmatrix} 6 & -2 & 9 \\ 2 & 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 7 & 1 & 2 \\ 6 & 1 & 5 \end{pmatrix}$$

Add or subtract corresponding elements:

$$\begin{pmatrix} 3 & 0 \\ -1 & 2 \\ 0 & 3 \end{pmatrix} - \begin{pmatrix} q & -3 \\ 1 & 1 \\ -4 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 3-q & 3 \\ -2 & 1 \\ 4 & 2 \end{pmatrix}$$

Scalar Multiplication

Solution 8

A scalar multiplies every element of a matrix, just as it does for a vector.

$$3\begin{pmatrix} 1 & 3 & -7 \\ 4 & 0 & 5 \end{pmatrix}$$

$$\begin{pmatrix} 3 & 9 & -21 \\ 12 & 0 & 15 \end{pmatrix}$$

$$A = \begin{pmatrix} q & -3 \\ 1 & 1 \\ -4 & 1 \end{pmatrix}$$

Find $2A$.

$$2A = \begin{pmatrix} 2q & -6 \\ 2 & 2 \\ -8 & 2 \end{pmatrix}$$

Further Example using algebraic terms

Solution 9

Find the values of k such that

$$\begin{pmatrix} -3 \\ k \end{pmatrix} + k \begin{pmatrix} 2k \\ 2k \end{pmatrix} = \begin{pmatrix} k \\ 6 \end{pmatrix}$$

First row

$2k^2 - k - 3 = (2k - 3)(k + 1) = 0$

$k = \frac{3}{2}$ or $k = -1$

Second row

$2k^2 + k - 6 = (2k - 3)(k + 2) = 0$

$k = \frac{3}{2}$ or $k = -2$

Conclusion

Both rows must hold. The common value is $k = \frac{3}{2}$.

Special Matrices

Solution 10

A **square matrix** has the same number of rows and columns.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

$$\begin{pmatrix} 3 & 1 & 4 \\ 2 & 2 & 5 \\ -3 & 4 & 3 \end{pmatrix}$$

An **identity matrix** has 1 on its leading diagonal and 0 elsewhere.

$$\mathbf{I}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\mathbf{I}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

A **zero matrix** has every element zero.

$$\mathbf{0} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Dimensions are usually clear from the context. Identity matrices are particularly useful in multiplication.

Practice: Solution 11

1. Find a, b, c given

$$\begin{pmatrix} 1 & 2 & 0 \\ a & b & c \end{pmatrix} + \begin{pmatrix} a & b & c \\ 1 & 2 & 0 \end{pmatrix} = \begin{pmatrix} c & 5 & c \\ c & c & c \end{pmatrix}$$

1

$2 + b = 5 \Rightarrow b = 3$

$b + 2 = c \Rightarrow c = 5$

$1 + a = c \Rightarrow a = 4$

2. Find a, b, c, d given

$$\begin{pmatrix} 5 & a \\ b & 0 \end{pmatrix} - 2\begin{pmatrix} c & 2 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 9 & 1 \\ 3 & d \end{pmatrix}$$

2

$5 - 2c = 9 \Rightarrow c = -2$

$a - 4 = 1 \Rightarrow a = 5$

$b - 2 = 3 \Rightarrow b = 5, \quad d = 2$

Multiply a Matrix by a vector Solution 12

Coffi Lab sells espresso for £3, cappuccino for £4 and tea for £2.

Rows below: espresso, cappuccino, tea.
Columns: Monday to Friday.

$$\begin{pmatrix} 13 & 9 & 7 & 15 & 1 \\ 8 & 7 & 4 & 6 & 2 \\ 6 & 4 & 0 & 3 & 5 \end{pmatrix}$$

How much did they make on Monday?

Monday

$13(3) + 8(4) + 6(2) = 39 + 32 + 12 = 83$

Monday revenue: £83.

Find the revenue for each day.

Daily revenue

Monday £83; Tuesday £63; Wednesday £37; Thursday £75; Friday £21.

Multiply a Matrix by a vector

Solution 13

You have calculated a row-vector × matrix product.

$$\begin{pmatrix}3 & 4 & 2\end{pmatrix}\begin{pmatrix}13 & 9 & 7 & 15 & 1\\ 8 & 7 & 4 & 6 & 2\\ 6 & 4 & 0 & 3 & 5\end{pmatrix} = \begin{pmatrix}83 & 63 & 37 & 75 & 21\end{pmatrix}$$

Suppose production costs are 50% of selling prices. Form a matrix containing both prices and costs.

Prices and costs

$$\begin{pmatrix}3 & 4 & 2\\ 1.50 & 2 & 1\end{pmatrix}$$

Multiply to find daily revenue and daily cost.

Product

$$\begin{pmatrix}3 & 4 & 2\\ 1.5 & 2 & 1\end{pmatrix}\begin{pmatrix}13 & 9 & 7 & 15 & 1\\ 8 & 7 & 4 & 6 & 2\\ 6 & 4 & 0 & 3 & 5\end{pmatrix} = \begin{pmatrix}83 & 63 & 37 & 75 & 21\\ 41.5 & 31.5 & 18.5 & 37.5 & 10.5\end{pmatrix}$$

Matrix Multiplication

Solution 14

Multiply each row of the left matrix by each column of the right matrix, adding the products.

$$\begin{pmatrix}1 & 0 & 3 & -2\\ 2 & 8 & 4 & 3\\ 7 & -1 & 0 & 2\end{pmatrix}\begin{pmatrix}5 & 1\\ 1 & 7\\ 0 & 3\\ 8 & -3\end{pmatrix}$$

First row, first column

$$1(5)+0(1)+3(0)+(-2)(8)=-11$$

Rest of first column

$$2(5)+8(1)+4(0)+3(8)=42$$

$$7(5)-1(1)+0(0)+2(8)=50$$

Complete product

$$\begin{pmatrix}-11 & 16\\ 42 & 61\\ 50 & -6\end{pmatrix}$$

Matrix Multiplication involving I

Solution 15

What do you notice?

$$\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}\begin{pmatrix}3 & 2\\1 & 0\end{pmatrix}$$

Left identity

$$\begin{pmatrix}3 & 2\\1 & 0\end{pmatrix}$$

$$\begin{pmatrix}3 & 2\\1 & 0\end{pmatrix}\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}$$

Right identity

$$\begin{pmatrix}3 & 2\\1 & 0\end{pmatrix}$$

For compatible identity matrices,

$\mathbf{AI} = \mathbf{IA} = \mathbf{A}$.

Multiplying by an identity matrix has no effect, just as multiplying a number by 1 has no effect.

The identity for ordinary addition is 0:
$a + 0 = 0 + a = a$.

Now you try..

Solution 16

a. $\begin{pmatrix}1 & 2\\3 & 4\end{pmatrix}\begin{pmatrix}2 & 0 & -1\\3 & 2 & 1\end{pmatrix}$

a

$$\begin{pmatrix}8 & 4 & 1\\18 & 8 & 1\end{pmatrix}$$

b. $\begin{pmatrix}1 & 2\\3 & 4\end{pmatrix}\begin{pmatrix}3\\-1\end{pmatrix}$

b

$$\begin{pmatrix}1\\5\end{pmatrix}$$

c. $\begin{pmatrix}1 & 2\\3 & 4\end{pmatrix}^2$

c

$$\begin{pmatrix}7 & 10\\15 & 22\end{pmatrix}$$

d. $\begin{pmatrix}1 & 2 & 3\end{pmatrix}\begin{pmatrix}1\\2\\3\end{pmatrix}$

d (14)

e. $\begin{pmatrix}1\\2\\3\end{pmatrix}\begin{pmatrix}1 & 2 & 3\end{pmatrix}$

e

$$\begin{pmatrix}1 & 2 & 3\\2 & 4 & 6\\3 & 6 & 9\end{pmatrix}$$

f. $\begin{pmatrix}1 & a\\0 & 1\end{pmatrix}^k$

f

$$\begin{pmatrix}1 & ak\\0 & 1\end{pmatrix}$$

Matrix Multiplication — Solution 17

The product AB exists only when the columns of A equal the rows of B.

$A_{m\times n}B_{n\times p} = (AB)_{m\times p}$

Only square matrices have positive integer powers greater than 1.

Find each product's dimensions, or state that it is invalid.

1. $2\times3 \times 3\times4$
2. $1\times3 \times 2\times3$
3. $6\times2 \times 2\times4$
4. $1\times3 \times 3\times1$

1
2×4

2
Not valid

3
6×4

4
1×1

5. $7\times5 \times 7\times5$
6. $10\times10 \times 10\times9$
7. $3\times3 \times 3\times3$

5
Not valid

6
10×9

7
3×3

Practice — Solution 18

3. Find in terms of a.

$$\begin{pmatrix}2 & a\\ 1 & -1\end{pmatrix}\begin{pmatrix}1 & 3 & 0\\ 0 & -1 & 2\end{pmatrix}$$

3
$$\begin{pmatrix}2 & 6-a & 2a\\ 1 & 4 & -2\end{pmatrix}$$

4. Find in terms of x.

$$\begin{pmatrix}3 & 2\\ -1 & x\end{pmatrix}\begin{pmatrix}x & -2\\ 1 & 3\end{pmatrix}$$

4
$$\begin{pmatrix}3x+2 & 0\\ 0 & 3x+2\end{pmatrix}$$

5. Given

$$A = \begin{pmatrix}1 & 2\\ 0 & 1\end{pmatrix}$$

find A^2, A^3 and suggest A^k.

5
$$A^2 = \begin{pmatrix}1 & 4\\ 0 & 1\end{pmatrix}$$
$$A^3 = \begin{pmatrix}1 & 6\\ 0 & 1\end{pmatrix}$$
$$A^k = \begin{pmatrix}1 & 2k\\ 0 & 1\end{pmatrix}$$

Practice — Solution 19

6. Given

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}$$

prove that $A^2 = 2A + 5I$.

6

$$A^2 = \begin{pmatrix} 7 & 4 \\ 6 & 7 \end{pmatrix}$$

$$2A + 5I = \begin{pmatrix} 2 & 4 \\ 6 & 2 \end{pmatrix} + \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix}$$

$$= \begin{pmatrix} 7 & 4 \\ 6 & 7 \end{pmatrix}$$

7. Given

$$A = \begin{pmatrix} p & 3 \\ 6 & p \end{pmatrix}, \quad B = \begin{pmatrix} q & 2 \\ 4 & q \end{pmatrix}$$

prove that $AB = BA$.

7

$$AB = \begin{pmatrix} pq + 12 & 2p + 3q \\ 6q + 4p & pq + 12 \end{pmatrix}$$

$$BA = \begin{pmatrix} pq + 12 & 3q + 2p \\ 4p + 6q & pq + 12 \end{pmatrix}$$

The corresponding entries agree, so $AB = BA$.

Matrix Multiplication on the CG50 — Solution 20

Matrix multiplication on a graphing calculator

Enter Run-Matrix mode and define two 2×2 matrices.

$$A = \begin{pmatrix} 3 & 2 \\ -1 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Enter each element in order. Return to the calculation screen, insert MatA and MatB using the matrix options menu, and evaluate MatA $\times$ MatB.

Button names vary by model.

Find AB, then check using the calculator.

Product

$$AB = \begin{pmatrix} 2 & 3 \\ 4 & -1 \end{pmatrix}$$

Matrix Multiplication on the Classwiz fx-991EX

Solution 21

Classwiz matrix practice

$$A = \begin{pmatrix} 2 & 3 \\ 0 & 2 \end{pmatrix},\ B = \begin{pmatrix} 4 & 0 \\ 1 & 1 \end{pmatrix}$$

$$C = \begin{pmatrix} 9 & 3 & 1 \\ 2 & 1 & 1 \end{pmatrix}$$

Open Matrix mode. Define MatA and MatB as 2×2, and MatC as 2×3. Enter each matrix. Use Matrix Calc and recall the stored matrices to evaluate the expressions.

Calculate

$4A,\ 3B - 2A,\ AB,\ BA,\ B^2,\ AC,\ BAC.$

a–e

$$4A = \begin{pmatrix} 8 & 12 \\ 0 & 8 \end{pmatrix},\quad 3B - 2A = \begin{pmatrix} 8 & -6 \\ 3 & -1 \end{pmatrix}$$

$$AB = \begin{pmatrix} 11 & 3 \\ 2 & 2 \end{pmatrix},\quad BA = \begin{pmatrix} 8 & 12 \\ 2 & 5 \end{pmatrix}$$

$$B^2 = \begin{pmatrix} 16 & 0 \\ 5 & 1 \end{pmatrix}$$

f–g

$$AC = \begin{pmatrix} 24 & 9 & 5 \\ 4 & 2 & 2 \end{pmatrix}$$

$$BAC = \begin{pmatrix} 96 & 36 & 20 \\ 28 & 11 & 7 \end{pmatrix}$$

Determinant of a matrix

Solution 22

A matrix can transform a point:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 7 \end{pmatrix}$$

Can an inverse transformation recover the original point?

$$\begin{pmatrix} ? & ? \\ ? & ? \end{pmatrix} \begin{pmatrix} 3 \\ 7 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

The required matrix is called the **inverse**. Just as the discriminant $b^2 - 4ac$ distinguishes quadratic root cases, the **determinant**, written $\det A$ or $|A|$, determines whether a square matrix has an inverse.

Determinants

Solution 23

$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$\det A = |A| = ad - bc.$

If $\det A = 0$, A is singular and has no inverse. Otherwise it is non-singular.

Find each determinant.

1. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
2. $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$
3. $\begin{pmatrix} 0 & 3 \\ -1 & -4 \end{pmatrix}$
4. $\begin{pmatrix} 10 & -2 \\ 4 & -1 \end{pmatrix}$

1
1

2
-2

3
3

4
-2

Test Your Understanding So Far

Solution 24

[Textbook] Given

$A = \begin{pmatrix} 4 & p+2 \\ -1 & 3-p \end{pmatrix}$

and that A is singular, find p.

Textbook

$4(3-p) + (p+2) = 0$

$14 - 3p = 0 \Rightarrow p = \frac{14}{3}$

Edexcel FP1 January 2010 Q5

Let $a \in \mathbb{R}$ and

$A = \begin{pmatrix} a & -5 \\ 2 & a+4 \end{pmatrix}$

a. Find $\det A$.

b. Show that A is non-singular for all a.

a

$\det A = a(a+4) + 10 = a^2 + 4a + 10$

b

$\det A = (a+2)^2 + 6 > 0$

Hence A is non-singular for every real a.

Determinants of 3× 3 matrices

Solution 25

Expand along the first row, with signs $+$, $-$, $+$.

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

$$= a\begin{vmatrix} e & f \\ h & i \end{vmatrix} - b\begin{vmatrix} d & f \\ g & i \end{vmatrix} + c\begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

Example

$$\det \begin{pmatrix} 3 & 1 & 4 \\ 2 & 2 & 5 \\ -3 & 4 & 3 \end{pmatrix}$$

Determinant

$3(6-20)-(6+15)+4(8+6)$

$= -42-21+56 = -7$

Test Your Understanding

Solution 26

1. Determine the determinant.

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 4 & 5 & -6 \\ -1 & 8 & 2 \end{pmatrix}$$

1

$\det A = (10+48)-2(8-6) = 54$

2. Given that the matrix is singular, find k.

$$A = \begin{pmatrix} 3 & k & 0 \\ -2 & 1 & 2 \\ 5 & 0 & k+3 \end{pmatrix}$$

2

$\det A = 3(k+3)-k[-2(k+3)-10]$

$= 2k^2+19k+9 = (2k+1)(k+9)$

$k = -\frac{1}{2}$ or -9

Minors

Solution 27

A minor is the determinant left after deleting the row and column of an entry.

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 4 & 5 & -6 \\ -1 & 8 & 2 \end{pmatrix}$$

Find the minor of

0

$$\begin{vmatrix} 4 & 5 \\ -1 & 8 \end{vmatrix} = 37$$

Find the minor of

-6

-6

$$\begin{vmatrix} 1 & 2 \\ -1 & 8 \end{vmatrix} = 10$$

Find the minor of

5

$$\begin{vmatrix} 1 & 0 \\ -1 & 2 \end{vmatrix} = 2$$

Exercise 6C Pure Core 1 - P106-108

Solution 28

2a. Find the value of a for which the matrix is singular.

$$\begin{pmatrix} a & 1+a \\ 3 & 2 \end{pmatrix}$$

a

$2a - 3(1+a) = 0$

$\Rightarrow a = -3$

2b. Find the value of a for which the matrix is singular.

$$\begin{pmatrix} 1+a & 3-a \\ a+2 & 1-a \end{pmatrix}$$

b

$(1+a)(1-a)$

$-(3-a)(a+2) = 0$

$-a - 5 = 0$

$a = -5$

2c. Find the value of a for which the matrix is singular.

$$\begin{pmatrix} 2+a & 1-a \\ 1-a & a \end{pmatrix}$$

c

$a(2+a) - (1-a)^2 = 0$

$\Rightarrow a = \frac{1}{4}$

3. Find the exact real values of k for which

$$M = \begin{pmatrix} -2 & 1-k \\ k-1 & k \end{pmatrix}$$

is singular.

Answer

$\det M = -2k - (1-k)(k-1)$

$= k^2 - 4k + 1$

$k^2 - 4k + 1 = 0$

$\Rightarrow k = 2 \pm \sqrt{3}$

4. Find the values of k for which

$$P = \begin{pmatrix} 3k & 4-k \\ k-2 & -k \end{pmatrix}$$

is singular.

Answer

$\det P = -3k^2 - (4-k)(k-2)$

$= -2k^2 - 6k + 8$

$2(k+4)(k-1) = 0$

$\Rightarrow k = -4$ or 1

8. Given

$$A = \begin{pmatrix} 2 & 1 & -4 \\ 2k+1 & 3 & k \\ 1 & 0 & 1 \end{pmatrix}$$

is singular, find k.

8

$\det A = 6 - (2k + 1 - k) - 4(-3)$

$= 17 - k$

$17 - k = 0$

$\Rightarrow k = 17$

11. Show that the matrix is singular for all a, b, c.

$$\begin{pmatrix} 0 & a & -b \\ -a & 0 & c \\ b & -c & 0 \end{pmatrix}$$

11

$\det A = 0 - a(-bc) - b(ac)$

$= abc - abc$

$= 0$

Therefore the matrix is always singular.

13. Find the determinant of

$$A = \begin{pmatrix} x-3 & -2 & 0 \\ 1 & x & -2 \\ -2 & -1 & x+1 \end{pmatrix}$$

and hence the values of x for which it is singular.

Determinant

$\det A = (x - 3)(x^2 + x - 2)$

$+ 2(x + 1 - 4)$

$= x^3 - 2x^2 - 3x$

$= x(x - 3)(x + 1)$

Singular values

$x = -1, 0, 3$

14. Given

$$M = \begin{pmatrix} 1 & -3 \\ 2 & 1 \end{pmatrix}, \quad N = \begin{pmatrix} -1 & k \\ 4 & 3 \end{pmatrix}$$

a. Find $\det M$.

b. Given $\det N = \det M$, find k.

c. Find MN.

d. Verify that $\det(MN) = \det M \det N$.

a

$\det M = 1 + 6 = 7$

b

$-3 - 4k = 7 \Rightarrow k = -\frac{5}{2}$

c

$$MN = \begin{pmatrix} -13 & -23/2 \\ 2 & -2 \end{pmatrix}$$

d

$\det(MN) = 26 + 23 = 49 = 7 \times 7$

The inverse satisfies $AA^{-1} = A^{-1}A = I$.

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

This exists only when $\det A \neq 0$.

To reverse $A\mathbf{x} = \mathbf{y}$, multiply on the left by A^{-1}:

$A^{-1}A\mathbf{x} = A^{-1}\mathbf{y}$

$\mathbf{x} = A^{-1}\mathbf{y}$

Practising the Inverse

Solution 35

Find the inverse, step by step.

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

First check that $ad - bc \neq 0$.

1: Swap

Swap the diagonal entries.

$$\begin{pmatrix} d & b \\ c & a \end{pmatrix}$$

2: Negate

Negate the other two entries.

$$\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

3: Divide

Divide by the determinant.

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Test Your Understanding

Solution 36

a. Find the inverse of

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

a

$$\frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

b. Find the inverse of

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

b

$$\frac{1}{2} \begin{pmatrix} -4 & 2 \\ 3 & -1 \end{pmatrix}$$

c. Find the inverse of

$$\begin{pmatrix} 7 & 2 \\ 1 & -3 \end{pmatrix}$$

c

$$\frac{1}{23} \begin{pmatrix} 3 & 2 \\ 1 & -7 \end{pmatrix}$$

d. Find the inverse of

$$\begin{pmatrix} 4 & p+2 \\ -1 & 3-p \end{pmatrix}$$

d

$$\frac{1}{14 - 3p} \begin{pmatrix} 3-p & -(p+2) \\ 1 & 4 \end{pmatrix}, \quad p \neq \frac{14}{3}$$

Matrix Proofs Involving Inverse

Solution 37

For non-singular matrices P, Q, show that $(PQ)^{-1} = Q^{-1}P^{-1}$.

$PQC = I \quad$ where $C = (PQ)^{-1}$

$QC = P^{-1} \Rightarrow C = Q^{-1}P^{-1}$

Given $BAB = I$, prove that $A = B^{-1}B^{-1}$.

$AB = B^{-1} \Rightarrow A = B^{-1}B^{-1}$

Multiply on the left first, then on the right. Matrix multiplication order matters.

Exercise 6D Pure Core 1 - P110-111

Solution 38

3a. For invertible square matrices, $ABC = I$. Prove that $B^{-1} = CA$.

a

$BC = A^{-1}$

$\Rightarrow B = A^{-1}C^{-1}$

$= (CA)^{-1}$

$B^{-1} = CA$

3b. Hence find B when

$$A = \begin{pmatrix} 0 & 1 \\ -1 & -6 \end{pmatrix}$$

$$C = \begin{pmatrix} 2 & 1 \\ -3 & -1 \end{pmatrix}$$

b

$CA = \begin{pmatrix} -1 & -4 \\ 1 & 3 \end{pmatrix}, \quad \det(CA)$

$= 1$

$B = \begin{pmatrix} 3 & 4 \\ -1 & -1 \end{pmatrix}$

7. Find A given

$$B = \begin{pmatrix} 5 & -4 \\ 2 & 1 \end{pmatrix}$$

$$AB = \begin{pmatrix} 11 & -1 \\ -8 & 9 \\ -2 & -1 \end{pmatrix}$$

Inverse

$$\det B = 13, \quad B^{-1} = \frac{1}{13}\begin{pmatrix} 1 & 4 \\ -2 & 5 \end{pmatrix}$$

A

$$A = (AB)B^{-1} = \frac{1}{13}\begin{pmatrix} 13 & 39 \\ -26 & 13 \\ 0 & -13 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 3 \\ -2 & 1 \\ 0 & -1 \end{pmatrix}$$

8. Let $a, b \neq 0$ and

$$A = \begin{pmatrix} 3a & b \\ 4a & 2b \end{pmatrix}$$

a. Find A^{-1}.

b. Given

$$B = \begin{pmatrix} -a & b \\ 3a & 2b \end{pmatrix}$$

and $B = XA$, find X.

a

$$\det A = 2ab, \quad A^{-1} = \frac{1}{2ab}\begin{pmatrix} 2b & -b \\ -4a & 3a \end{pmatrix}$$

b

$$X = BA^{-1} = \frac{1}{2ab}\begin{pmatrix} -6ab & 4ab \\ -2ab & 3ab \end{pmatrix}$$

$$= \begin{pmatrix} -3 & 2 \\ -1 & 3/2 \end{pmatrix}$$

Exercise 6D Pure Core 1 - P110-111 — Solution 41

10. Let

$$M = \begin{pmatrix} 2 & 3 \\ k & -1 \end{pmatrix}$$

a. For which k does the inverse exist?

b. Find the inverse.

a

$$\det M = -2 - 3k \neq 0 \Rightarrow k \neq -\frac{2}{3}$$

b

$$M^{-1} = \frac{1}{3k+2}\begin{pmatrix} 1 & 3 \\ k & -2 \end{pmatrix}$$

Exercise 6D Pure Core 1 - P110-111 — Solution 42

13. Let

$$A = \begin{pmatrix} a & 2 \\ 3 & 2a \end{pmatrix}$$

a. Find A^{-1}.

b. Find the two values of a for which no inverse exists.

a

$$\det A = 2a^2 - 6$$

$$A^{-1} = \frac{1}{2a^2-6}\begin{pmatrix} 2a & -2 \\ -3 & a \end{pmatrix}$$

b

$$2a^2 - 6 = 0 \Rightarrow a = \pm\sqrt{3}$$

Proof for the Formula of the Inverse of a 2x2 Matrix

Solution 43

For interest: derive the inverse by row operations. Let $\Delta = ad - bc \neq 0$. For $a \neq 0$, start with

$$\left[\begin{array}{cc|cc} a & b & 1 & 0 \\ c & d & 0 & 1 \end{array}\right]$$

Divide row 1 by a, then subtract c times row 1 from row 2:

$$\left[\begin{array}{cc|cc} 1 & b/a & 1/a & 0 \\ 0 & \Delta/a & -c/a & 1 \end{array}\right]$$

Multiply row 2 by a/Δ, then subtract b/a times row 2 from row 1:

$$\left[\begin{array}{cc|cc} 1 & 0 & d/\Delta & -b/\Delta \\ 0 & 1 & -c/\Delta & a/\Delta \end{array}\right]$$

$$M^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

If $a = 0$, interchange the rows first. The same formula also follows by multiplying the proposed inverse by M.

Matrix Transpose

Solution 44

The transpose A^T interchanges rows and columns.

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

An $m \times n$ matrix becomes $n \times m$.

Beyond the exam: transposition moves a transformation between the two vectors in a dot product.

$$(A\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (A^T\mathbf{v})$$

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 7 \\ 8 \end{pmatrix}$$

$$\begin{pmatrix} 17 \\ 39 \end{pmatrix} \cdot \begin{pmatrix} 7 \\ 8 \end{pmatrix} = 431$$

$$\begin{pmatrix} 5 \\ 6 \end{pmatrix} \cdot \begin{pmatrix} 31 \\ 46 \end{pmatrix} = 431$$

Inversing a 3× 3 matrix

Solution 45

General inverse method, illustrated with

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

1. Find the determinant.
2. Find each minor by deleting its row and column.
3. Apply alternating signs, beginning with $+$ at the top left, to form the cofactor matrix C.
4. Transpose C and divide by the determinant.

1–2

$$\det A = 4 - 6 = -2$$

$$M = \begin{pmatrix} 4 & 3 \\ 2 & 1 \end{pmatrix}$$

3–4

$$C = \begin{pmatrix} 4 & -3 \\ -2 & 1 \end{pmatrix}$$

$$A^{-1} = \frac{C^T}{\det A} = -\frac{1}{2}\begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix}$$

Inversing a 3× 3 matrix

Solution 46

Find the inverse of

$$A = \begin{pmatrix} 1 & 3 & 1 \\ 0 & 4 & 1 \\ 2 & -1 & 0 \end{pmatrix}$$

1: Determinant

$$\det A = 1 - 3(-2) - 8 = -1$$

2: Minors

$$M = \begin{pmatrix} 1 & -2 & -8 \\ 1 & -2 & -7 \\ -1 & 1 & 4 \end{pmatrix}$$

3: Cofactors

$$C = \begin{pmatrix} 1 & 2 & -8 \\ -1 & -2 & 7 \\ -1 & -1 & 4 \end{pmatrix}$$

4: Inverse

$$A^{-1} = \frac{C^T}{-1} = \begin{pmatrix} -1 & 1 & 1 \\ -2 & 2 & 1 \\ 8 & -7 & -4 \end{pmatrix}$$

Doing with your Classwiz

Solution 47

Use your Classwiz to invert

$$A = \begin{pmatrix} 1 & 3 & 1 \\ 0 & 4 & 1 \\ 2 & -1 & 0 \end{pmatrix}$$

1. Open Matrix mode and select MatA.
2. Choose 3 rows and 3 columns, then enter the numbers, pressing $=$ after each.
3. Press AC to start a calculation.
4. Use OPTN to insert MatA.
5. Use the dedicated x^{-1} key, not the general power key.
6. *Press* $=$.

Menu names depend on the calculator model.

Further Example

Solution 48

Given

$$A = \begin{pmatrix} -2 & 3 & -3 \\ 0 & 1 & 0 \\ 1 & -1 & 2 \end{pmatrix}$$

$$(AB)^{-1} = \begin{pmatrix} 8 & -17 & 9 \\ -5 & 10 & -6 \\ -3 & 5 & -4 \end{pmatrix}$$

a. Show $A^{-1} = A$.

b. Find B^{-1}.

a

$$A^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I$$

Hence $A^{-1} = A$.

b

$$(AB)^{-1} = B^{-1}A^{-1}$$

$$B^{-1} = (AB)^{-1}A = \begin{pmatrix} -7 & -2 & -6 \\ 4 & 1 & 3 \\ 2 & 0 & 1 \end{pmatrix}$$

Test Your Understanding

Solution 49

June 2011 Q7:

$$M = \begin{pmatrix} k & -1 & 1 \\ 1 & 0 & -1 \\ 3 & -2 & 1 \end{pmatrix}$$

where $k \neq 1$.

a. Show $\det M = 2 - 2k$.

b. Find M^{-1}.

a

$$\det M = -2k + (1 + 3) - 2 = 2 - 2k$$

b

$$M^{-1} = \frac{1}{2 - 2k} \begin{pmatrix} -2 & -1 & 1 \\ -4 & k - 3 & k + 1 \\ -2 & 2k - 3 & 1 \end{pmatrix}$$

Exercise 6E Pure Core 1 - P115-116

Solution 50

1. Use your calculator to find the inverse; the cofactor method is shown for comparison.

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

Determinant and minors

$$\det A = 3, \quad M = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

Cofactors

$$C = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix} = C^T$$

Inverse

$$A^{-1} = \frac{1}{3} \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$$

Exercise 6E Pure Core 1 - P115-116 Solution 51

4. Let

$$A = \begin{pmatrix} 2 & 0 & 3 \\ k & 1 & 1 \\ 1 & 1 & 4 \end{pmatrix}$$

a. Show $\det A = 3(k+1)$.

b. For $k \neq -1$, find A^{-1}.

a

$\det A = 2(4-1) + 3(k-1) = 3(k+1)$

b: Minors and cofactors

$$M = \begin{pmatrix} 3 & 4k-1 & k-1 \\ -3 & 5 & 2 \\ -3 & 2-3k & 2 \end{pmatrix}$$

$$C = \begin{pmatrix} 3 & 1-4k & k-1 \\ 3 & 5 & -2 \\ -3 & 3k-2 & 2 \end{pmatrix}$$

b: Inverse

$$A^{-1} = \frac{1}{3(k+1)} \begin{pmatrix} 3 & 3 & -3 \\ 1-4k & 5 & 3k-2 \\ k-1 & -2 & 2 \end{pmatrix}$$

Exercise 6E Pure Core 1 - P115-116 Solution 52

5. Given

$$A = \begin{pmatrix} 5 & a & 4 \\ b & -7 & 8 \\ 2 & -2 & c \end{pmatrix}$$

and $A = A^{-1}$, find a, b, c.

Constants

Use $A^2 = I$ and compare entries.

$(A^2)_{12} = -2a - 8 = 0 \Rightarrow a = -4$

$(A^2)_{21} = 16 - 2b = 0 \Rightarrow b = 8$

$(A^2)_{31} = -2b + 2c + 10 = 0 \Rightarrow c = 3$

Substitution gives $A^2 = I$, checking all entries.

Exercise 6E Pure Core 1 - P115-116 — Solution 53

9. Let

$$M = \begin{pmatrix} 2 & k & 3 \\ -1 & 2 & 1 \\ 1 & -1 & -1 \end{pmatrix}$$

a. For which k does M^{-1} exist?

b. Find M^{-1}.

a

$\det M = 2(-1) - k(0) + 3(-1) = -5$

An inverse exists for every real k.

b: Cofactors

$$C = \begin{pmatrix} -1 & 0 & -1 \\ k-3 & -5 & k+2 \\ k-6 & -5 & k+4 \end{pmatrix}$$

b: Inverse

$$M^{-1} = \frac{1}{5}\begin{pmatrix} 1 & 3-k & 6-k \\ 0 & 5 & 5 \\ 1 & -k-2 & -k-4 \end{pmatrix}$$

Exercise 6E Pure Core 1 - P115-116 — Solution 54

10. Given that

$$A = \begin{pmatrix} p & 2p & 3 \\ 4 & -1 & 1 \\ 1 & -2 & 0 \end{pmatrix}$$

is non-singular, find A^{-1}.

Determinant

$\det A = 2p + 2p - 21 = 4p - 21$

$p \neq \frac{21}{4}$

Cofactors

$$C = \begin{pmatrix} 2 & 1 & -7 \\ -6 & -3 & 4p \\ 2p+3 & 12-p & -9p \end{pmatrix}$$

Inverse

$$A^{-1} = \frac{1}{4p-21}\begin{pmatrix} 2 & -6 & 2p+3 \\ 1 & -3 & 12-p \\ -7 & 4p & -9p \end{pmatrix}$$

Using Matrices For Simultaneous Equations

Solution 55

Solve the simultaneous equations using an inverse matrix.

$$-x + 6y - 2z = 21$$
$$6x - 2y - z = -16$$
$$-2x + 3y + 5z = 24$$

1: Matrix equation

$$\begin{pmatrix} -1 & 6 & -2 \\ 6 & -2 & -1 \\ -2 & 3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 21 \\ -16 \\ 24 \end{pmatrix}$$

2: Inverse

$$A^{-1} = \frac{1}{189} \begin{pmatrix} 7 & 36 & 10 \\ 28 & 9 & 13 \\ -14 & 9 & 34 \end{pmatrix}$$

3: Solution

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1} \begin{pmatrix} 21 \\ -16 \\ 24 \end{pmatrix} = \begin{pmatrix} -1 \\ 4 \\ 2 \end{pmatrix}$$

Check by substitution or your calculator's simultaneous-equation solver.

Forming the equations yourself

Solution 56

A population of 1000 mole-rats has 100 more females than males. After one year, the numbers of males and females rise by 2% and 3% respectively; the number of young falls by 4%. The total population falls by 20. Find the initial numbers.

1: Model

Let x, y, z be the numbers of males, females and young.

$$x + y + z = 1000$$
$$x - y = -100$$
$$1.02x + 1.03y + 0.96z = 980$$

2: Invert

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1.02 & 1.03 & 0.96 \end{pmatrix}$$

$$A^{-1} = \frac{1}{13} \begin{pmatrix} -96 & 7 & 100 \\ -96 & -6 & 100 \\ 205 & -1 & -200 \end{pmatrix}$$

3: Population

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1} \begin{pmatrix} 1000 \\ -100 \\ 980 \end{pmatrix} = \begin{pmatrix} 100 \\ 200 \\ 700 \end{pmatrix}$$

Consistency of linear equations

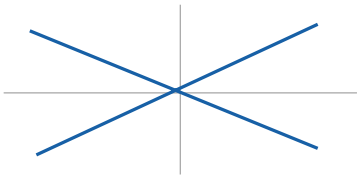
Solution 57

A system is consistent if it has at least one solution. In two dimensions, solutions are common points of the lines.

1. One intersection: unique solution.

   $x + 2y = 1$

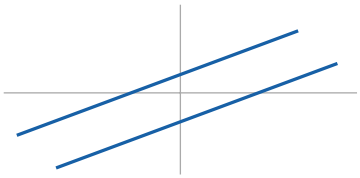
   $3x - y = 0$



2. Distinct parallel lines: no solution.

   $x - 3y = 1$

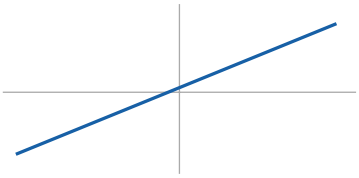
   $x - 3y = 2$



3. Coincident lines: infinitely many solutions.

   $x - 3y = 1$

   $2x - 6y = 2$

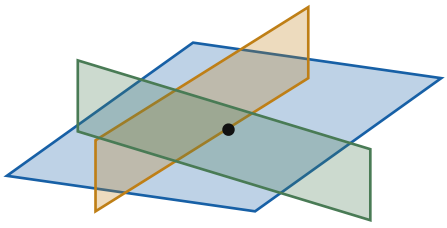


Extending consistency to 3 variables

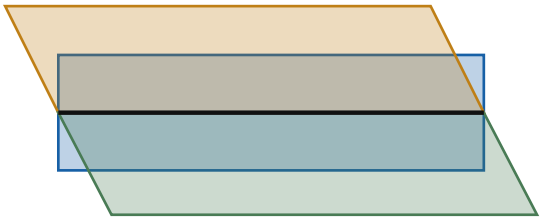
Solution 58

In three dimensions, $ax + by + cz = d$ describes a plane. A simultaneous solution belongs to all three planes.

1. One common point: unique solution.



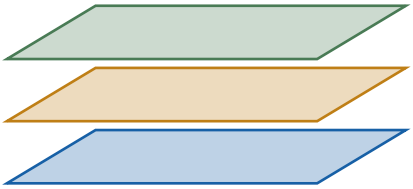
2. One common line (a sheaf): infinitely many solutions.



3. Prism arrangement: pairwise intersections, but no common point. Inconsistent.



4. Distinct parallel planes: no common point. Inconsistent.



Extending consistency to 3 variables

Solution 59

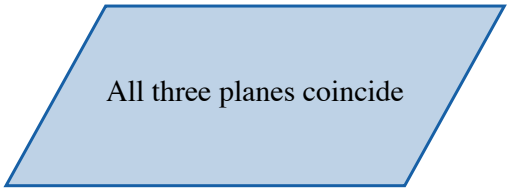
5. Coincident planes

If all three equations describe the same plane, every point on that plane is a solution.

$x + 2y - z = 3$

$2x + 4y - 2z = 6$

$-x - 2y + z = -3$



All three equations are multiples of one another.

The system is consistent with infinitely many solutions.

A zero determinant alone does not tell us whether there are no solutions or infinitely many: we must inspect the equations.

Example

Solution 60

Describe the intersections when $k = 0, 1, -6$.

$3x - ky - 6z = k$

$kx + 3y + 3z = 2$

$-3x - y + 3z = -2$

First take $k = 0$.

Determinant

$$A = \begin{pmatrix} 3 & 0 & -6 \\ 0 & 3 & 3 \\ -3 & -1 & 3 \end{pmatrix}$$

$\det A = -18 \neq 0$

Intersection

The system has a unique solution: the planes meet at one point.

$(x, y, z) = \left(\frac{4}{3}, 0, \frac{2}{3}\right)$

Example

Solution 61

For $k = 1$:

$$3x - y - 6z = 1$$
$$x + 3y + 3z = 2$$
$$-3x - y + 3z = -2$$

k = 1

The determinant is zero. Eliminating z gives

$5x + 5y = 5, \qquad 4x + 4y = 4$

These agree: infinitely many solutions on a common line (a sheaf).

For $k = -6$:

$$3x + 6y - 6z = -6$$
$$-6x + 3y + 3z = 2$$
$$-3x - y + 3z = -2$$

k = −6

Taking $E_1 - E_2 + 3E_3$ gives

$0 = -14$

No common point: the system is inconsistent.

Test Your Understanding

Solution 62

Find the values of k for which the system has a unique solution.

$$2x + 3y - z = 13$$
$$3x - y + kz = 11$$
$$x + y + z = 7$$

Determinant

$$A = \begin{pmatrix} 2 & 3 & -1 \\ 3 & -1 & k \\ 1 & 1 & 1 \end{pmatrix}$$

$\det A = 2(-1 - k) - 3(3 - k) - 4$

$= k - 15$

Condition

A unique solution exists exactly when

$k \neq 15$

Exercise 6F Pure Core 1 - P120-121

Solution 63

2. Solve the system and show that it has a unique solution.

$$x - 3y - 4z = 3$$
$$6x + 5y - 7z = 30$$
$$x + 4y + 6z = -3$$

Matrix

$$A = \begin{pmatrix} 1 & -3 & -4 \\ 6 & 5 & -7 \\ 1 & 4 & 6 \end{pmatrix}$$

$\det A = 111 \neq 0$

Inverse

$$A^{-1} = \frac{1}{111}\begin{pmatrix} 58 & 2 & 41 \\ -43 & 10 & -17 \\ 19 & -7 & 23 \end{pmatrix}$$

Solution

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1}\begin{pmatrix} 3 \\ 30 \\ -3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$$

The non-zero determinant ensures this solution is unique.

Exercise 6F Pure Core 1 - P120-121

Solution 64

3. Phyllis invests £3000 in funds A, B and C, putting £190 more in A than B. After two years their values change by $+1\%$, $+2.5\%$ and -1.5% respectively. Her total investment increases by £41. Find the original investments.

Model

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1.01 & 1.025 & 0.985 \end{pmatrix}\begin{pmatrix} A \\ B \\ C \end{pmatrix} = \begin{pmatrix} 3000 \\ 190 \\ 3041 \end{pmatrix}$$

Inverse

$$M^{-1} = \frac{1}{0.065}\begin{pmatrix} -0.985 & 0.04 & 1 \\ -0.985 & -0.025 & 1 \\ 2.035 & -0.015 & -2 \end{pmatrix}$$

Answer

$$\begin{pmatrix} A \\ B \\ C \end{pmatrix} = \frac{1}{0.065}\begin{pmatrix} 93.6 \\ 81.25 \\ 20.15 \end{pmatrix} = \begin{pmatrix} 1440 \\ 1250 \\ 310 \end{pmatrix}$$

Investments: £1440, £1250 and £310.

5. These planes do not intersect at a single point. Find a and describe their intersection.

$$x + ay + 2z = a$$
$$x - y - z = a$$
$$x + 4y + 4z = 0$$

Parameter

$$\det \begin{pmatrix} 1 & a & 2 \\ 1 & -1 & -1 \\ 1 & 4 & 4 \end{pmatrix} = -5(a-2) = 0$$

$$a = 2$$

Intersection

With $a = 2$, subtract equation 2 from equation 1:

$$3(y+z) = 0 \Rightarrow y + z = 0$$

Subtract equation 2 from equation 3:

$$5(y+z) = -2$$

This is a contradiction. The planes form a prism arrangement: no common point.

6. Given

$$M = \begin{pmatrix} 1 & 4 & q \\ 2 & 3 & -3 \\ q & q & -2 \end{pmatrix}$$

a. Show that singularity implies $q^2 + 9q - 10 = 0$.

b. Classify the system $M\mathbf{x} = (-16, q/2, -2)^T$ for $q = -10, 2, 1$.

a

$$\det M = -q^2 - 9q + 10$$

$$\det M = 0 \iff (q+10)(q-1) = 0$$

b: q = −10

$q = -10$: infinitely many solutions,

$$(x, y, z) = \left(\frac{28 - 18t}{5}, \frac{17t - 27}{5}, t\right)$$

a sheaf of planes.

b: q = 2

$q = 2$: $\det M = -12 \neq 0$. One point,

$$(x, y, z) = (-4, -1, -4)$$

b: q = 1

$q = 1$: elimination gives a contradiction, so no solution (a prism arrangement).